

can be calculated by multiplying the efficiencies of the blocks in the system:

$$\text{System eff.} = (\text{Solar eff.}) \times (\text{MPP eff.}) \times (\text{battery charge eff.}) \times (\text{switcher eff.})$$

Assuming that the solar cells are 25% efficient, the MPP is 85%, the battery can charge at 95% and the switcher is 85%, this means that 17% of the energy falling on the solar cells will actually reach the system load. Even if we disregard the 25% efficiency of the solar cells, over which we have little control, only 69% of the power from the solar cells reaches the load to do useful work. The other 31% of the energy is lost along the way, as heat, in the various modules.

The system also has another problem; part of the battery's function is to act as a shunt regulator for the system. If the battery fails, the voltage at the input of the load's switching regulator can vary significantly. If the battery fails open, then the voltage at the input of the load's switching regulator will vary up and down—dependent upon system load and the amount of sun on the solar cells. If the battery fails shorted, then the system is completely dead and the MPP just dumps the energy into the shorted battery.

One solution to these problems is to redesign the MPP to supply a constant output bus voltage that is slightly higher than the fully charged voltage of the battery. The battery is then charged from this bus voltage through a switcher-based charger, and the battery can drive the bus through a diode if the bus voltage sags due to heavy load or low light conditions. The diode also isolates the battery from the bus, when the MPP is driving the bus in full light conditions. Using this system solves the variable output voltage problem of the previous MPP design, and the diode automatically connects the battery to the bus, to take up the slack when the MPP is incapable of supplying energy to the load. Note, to completely alleviate the shorting problem, the charger will need an active disconnect to isolate

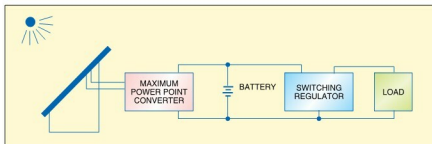


Fig. 1: Solar array with MPP converter

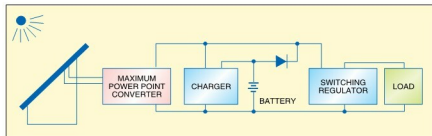


Fig. 2: Solar array with isolated battery and charger

the battery in the event of a short. (See Fig. 2.)

So, we now have a system that will survive a battery failure. In fact, we can put multiple batteries on the bus, as long as each is isolated by a diode and charged by separate chargers. But, this has moved the excess heat problem back to the solar array. More importantly, what has this done to our efficiency?

Well, let's start with the efficiency problem. If the purpose of the system is charging a battery to run night lights, then we have just introduced two new losses into the system; the converter efficiency of the battery charger, and the loss in the diode when we operate from the battery:

$$\text{System eff. night} = (\text{MPP eff.}) \times (\text{charger eff.}) \times (\text{battery charge eff.}) \times (\text{switcher}) - \text{diode loss}$$

Assuming a 10V system that is drawing 1 amp from the battery, the diode loss is approximately $\frac{1}{2}$ a watt or 5% during discharge. We don't have a load during the day, other than charging, so we have the stacked efficiency of the MPP, the charger and the charging efficiency of the battery while charging, which is the 69% result shown in the previous calculation. The

additional 5%, due to the diode drops combined with the 85% efficiency of the switching regulator, drops the overall efficiency of the system to 55%. Not particularly helpful as far as efficiency, but it does isolate battery failures from the system.

What about a system that is active both during the day and at night? If we use the system described above, we only get 55% at night. However, during the day we only have the MPP and the load's switching regulator between the solar cells and the load, so our efficiency improves to about 72%:

$$\text{System eff. daylight} = (\text{MPP eff.}) \times (\text{switcher eff.}); \text{ neglecting solar cells efficiency}$$

If the day and night are equal length, then the overall efficiency for the system, through the full day, is the average of the two efficiencies, or 64%. A little less than our original efficiency of 69%, but the system is now able to continue operation—even with a battery failure (possibly with several battery failures). And, if the day is longer than the night, our efficiency will climb higher toward the 72% range. For example, if we assume a satellite with a 10-hour orbit, which spends only two